

Chapter 3

Metals and Non-Metals

1. Which of the following is the most ductile metal?

- A) Silver
- B) Gold
- C) Copper
- D) Iron

Answer: B) Gold

2. The property of metals by which they can be drawn into thin wires is called:

- A) Malleability
- B) Ductility
- C) Sonority
- D) Conductivity

Answer: B) Ductility

3. Which of the following metals is a liquid at room temperature?

- A) Sodium
- B) Mercury
- C) Aluminium
- D) Zinc

Answer: B) Mercury

4. Which of the following is a non-metal that is lustrous?

- A) Sulphur
- B) Iodine
- C) Phosphorus
- D) Carbon

Answer: B) Iodine

5. Which of the following metals is stored under kerosene?

- A) Aluminium
- B) Calcium
- C) Sodium
- D) Iron

Answer: C) Sodium

6. Which non-metal is essential for breathing?

- A) Hydrogen
- B) Carbon dioxide
- C) Oxygen
- D) Nitrogen

Answer: C) Oxygen

7. Which of the following metals does not react with cold or hot water but reacts with steam?

- A) Sodium
- B) Magnesium
- C) Calcium
- D) Aluminium

Answer: B) Magnesium

8. Which gas is released when a metal reacts with an acid?

- A) Oxygen
- B) Hydrogen
- C) Nitrogen
- D) Carbon dioxide

Answer: B) Hydrogen

9. Which of the following metals will displace hydrogen from dilute acids?

- A) Copper
- B) Silver
- C) Gold
- D) Zinc

Answer: D) Zinc

10. The metal which does not corrode easily is:

- A) Iron
- B) Silver
- C) Aluminium
- D) Gold

Answer: D) Gold

11. The metal used for making electric wires is:

- A) Silver
- B) Iron
- C) Copper

D) Aluminium

Answer: C) Copper

12. Which of the following is the most reactive metal?

A) Copper

B) Iron

C) Sodium

D) Silver

Answer: C) Sodium

13. Which of the following is a noble metal?

A) Iron

B) Copper

C) Silver

D) Zinc

Answer: C) Silver

14. Non-metals react with oxygen to form:

A) Basic oxides

B) Acidic oxides

C) Neutral oxides

D) Amphoteric oxides

Answer: B) Acidic oxides

15. Which metal is used in thermite welding?

A) Iron

B) Aluminium

C) Copper

D) Zinc

Answer: B) Aluminium

16. The reactivity of metals can be compared using the:

A) Atomic table

B) Activity series

C) Periodic series

D) Ionisation chart

Answer: B) Activity series

17. Which of the following oxides is amphoteric in nature?

- A) Na_2O
- B) Al_2O_3
- C) MgO
- D) SO_2

Answer: B) Al_2O_3

18. Which of the following metals forms a protective oxide layer and prevents further corrosion?

- A) Iron
- B) Aluminium
- C) Copper
- D) Silver

Answer: B) Aluminium

19. The non-metal which is a good conductor of electricity is:

- A) Sulphur
- B) Graphite
- C) Phosphorus
- D) Iodine

Answer: B) Graphite

20. The process of coating iron with zinc to prevent rusting is called:

- A) Electroplating
- B) Rusting
- C) Galvanisation
- D) Refining

Answer: C) Galvanisation

21. Which of the following metals is used in thermometers?

- A) Sodium
- B) Mercury
- C) Iron
- D) Zinc

Answer: B) Mercury

22. The metal that reacts vigorously with water producing hydrogen gas is:

- A) Aluminium
- B) Calcium
- C) Sodium

D) Zinc

Answer: C) Sodium

23. Which of the following is not a metal?

A) Magnesium

B) Phosphorus

C) Zinc

D) Copper

Answer: B) Phosphorus

24. Which of the following pairs of elements can form an ionic bond?

A) Sodium and chlorine

B) Carbon and hydrogen

C) Hydrogen and oxygen

D) Nitrogen and oxygen

Answer: A) Sodium and chlorine

25. Which metal is purified by electrolytic refining?

A) Sodium

B) Aluminium

C) Copper

D) Calcium

Answer: C) Copper

26. Which of the following non-metals exists in liquid form?

A) Bromine

B) Iodine

C) Carbon

D) Sulphur

Answer: A) Bromine

27. The metal which can be cut with a knife is:

A) Aluminium

B) Zinc

C) Sodium

D) Copper

Answer: C) Sodium

28. The least reactive metal among the following is:

- A) Zinc
- B) Copper
- C) Iron
- D) Sodium

Answer: B) Copper

29. The non-metal which forms a neutral oxide is:

- A) Carbon
- B) Nitrogen
- C) Oxygen
- D) Hydrogen

Answer: C) Oxygen

30. Aqua regia is a mixture of:

- A) HCl and H_2SO_4
- B) HCl and HNO_3 (3:1)
- C) HNO_3 and H_2SO_4
- D) H_2SO_4 and H_3PO_4

Answer: B) HCl and HNO_3 (3:1)
